Double Integrals

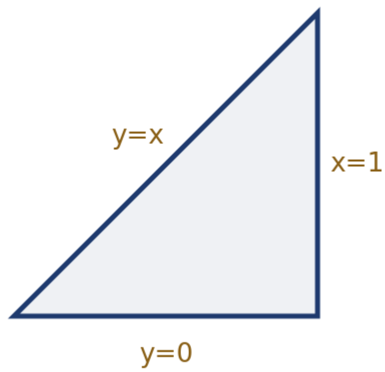
Iterated integrals over rectangles

1 Evaluate $\int_0^2 \int_0^3 (x^2 + y)\, dy\, dx$.

2 Evaluate $\int_1^2 \int_0^1 6xy^2\, dx\, dy$.

Double integrals over general regions

3 Evaluate $\int_0^1 \int_0^x (x + y)\, dy\, dx$, integrating over the triangular region $0 \le y \le x \le 1$.



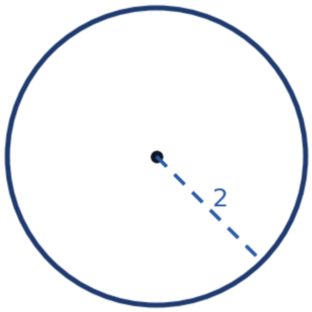
4 Use a double integral to find the volume under $f(x, y) = 4 - x - y$ over the rectangle $0 \le x \le 1$, $0 \le y \le 1$.

Reversing the order of integration

5 Evaluate $\int_0^1 \int_y^1 e^{x^2}\, dx\, dy$ by first reversing the order of integration (the region is $0 \le y \le x \le 1$).

Double integrals in polar coordinates

6 Evaluate $\iint_D (x^2 + y^2)\, dA$ where D is the disk $x^2 + y^2 \le 4$, by converting to polar coordinates ($dA = r\, dr\, d\theta$).



7 Find the area of the region enclosed by $r = 2\cos\theta$ using a double integral in polar coordinates.

Average value over a region

8 Find the average value of $f(x, y) = xy$ over the rectangle $0 \le x \le 2$, $0 \le y \le 3$ (average value $= \frac{1}{A}\iint_D f\, dA$, where A is the region's area).

Mass and center of a region

9 A lamina occupies the rectangle $0 \le x \le 2$, $0 \le y \le 1$ with density $\rho(x, y) = x + y$. Find its total mass, $M = \iint_D \rho\, dA$.

10 For a lamina occupying $0 \le x \le 1$, $0 \le y \le 1$ with constant density $\rho = 1$, find the x-coordinate of the center of mass, $\bar{x} = \frac{1}{M}\iint_D x\, dA$ (where M is the total mass).

Triple integral setup (a first look)

11 Set up (do not evaluate) the triple integral for the volume of the solid bounded by $0 \le x \le 1$, $0 \le y \le 2$, $0 \le z \le 3$.

Double integrals with a variable upper limit

12 Evaluate $\int_0^1 \int_0^{y^2} 3x\, dx\, dy$, integrating over the region $0 \le x \le y^2$, $0 \le y \le 1$.

13 Set up and evaluate the double integral for the volume under $f(x, y) = xy^2$ over the triangular region bounded by $y = x$, $y = 0$, and $x = 2$.

An applied double-integral problem

14 A metal plate occupies the region $0 \le x \le 3$, $0 \le y \le 2$, with temperature distribution $T(x, y) = 50 - x^2 - y^2$ (degrees Celsius). Find the average temperature over the plate.

Volume between two surfaces

15 Find the volume of the solid between $z = 8 - x^2 - y^2$ and $z = 0$ over the region where $z \ge 0$ requires $x^2 + y^2 \le 8$, using polar coordinates.

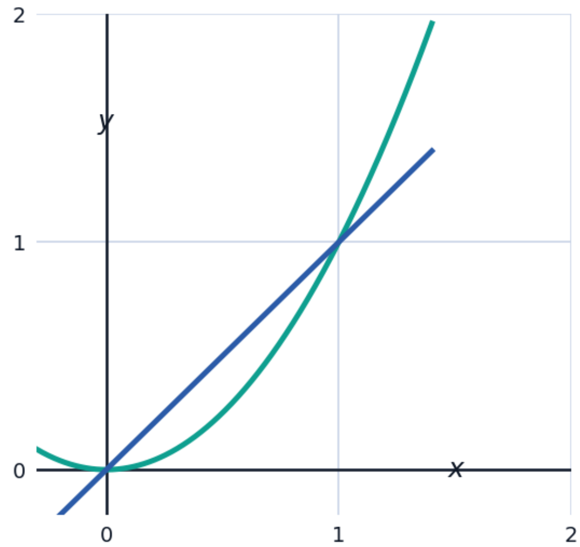
Double integrals with exponential integrands

16 Evaluate $\int_0^1 \int_0^1 xe^{xy}\,dy\,dx$ by integrating with respect to y first.

A capstone applied double-integral problem

17 An architect is designing a swimming pool whose floor depth varies according to $d(x, y) = 1 + 0.2x + 0.1y$ (meters) over a rectangular footprint $0 \le x \le 10$, $0 \le y \le 5$ (meters). Set up and evaluate the double integral for the total volume of water needed to fill the pool, then find the average depth by dividing by the pool's surface area.

18 For the double integral $\iint_D (2x + y)\,dA$ where D is the region bounded by $y = x^2$ and $y = x$, complete each step:



a Find the intersection points of the two curves to determine the bounds.

b Set up the iterated integral with x as the outer variable.

c Evaluate the integral.